Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____S0/0/0________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ____129_____ 2

Does R2 show up as an OSPF neighbor on R1? ___YES_____ 2

Does R2 show up as an OSPF neighbor on R3? ____NO____ 2

What does this information tell you?

All traffic destined for the 192.168.2.0/24 network from R3 will be forwarded via R1. The S0/0/1 interface on R2 remains configured as a passive interface, meaning OSPF routing updates are not being advertised through it. The total accumulated cost of 129 is due to the path from R3 to the 192.168.2.0/24 network requiring two T1 (1.544 Mb/s) serial links (each with a cost of 64) and the R2 Gigabit 0/0 LAN link (with a cost of 1).

2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ______S0/0/1_______ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65; 64(one T1 serial link) + 1(R2 Gigabit 0/0 LAN link) = 65. 2

Is R2 listed as an OSPF neighbor to R3? ____YES____ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because modern equipment supports link speeds exceeding 100 Mb/s, the default OSPF reference-bandwidth must be increased to calculate path costs more accurately for these faster interfaces.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 (781+1=782).

The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 (781+64=845).

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link has a cost of 781, and the route to the 192.168.23.0/24 network travels over 2 serial links. Thus, 781 * 2 = 1,562.

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because OSPF will choose the route with the smallest accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0 (781 + 781 + 1 = 1,563). This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0 (1565 + 1 = 1,566).

4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Controlling the router ID is essential for OSPF stability, as it directly influences DR/BDR elections on multi-access networks. Since a router ID derived from a physical interface may change if that interface fails, it should be set manually using either a loopback interface or the router-id command to ensure consistency.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   Because the DR/BDR election process is only an issue on a multi-access network such as Ethernet or Frame Relay. The serial links used in this lab are point-to-point links, thus, no DR/BDR election is performed.

   4

3. Why would you want to set an OSPF interface to passive?

   Configuring a LAN interface as passive in OSPF prevents routing protocol updates from being sent out that interface, which conserves bandwidth. Importantly, the router continues to advertise the connected network to its OSPF neighbors.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of areas. Routing information is contained within an area, and only summary information is passed between areas (via Area 0), which reduces the size of the routing table and the impact of topology changes.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The RIP route. Routers use the Administrative Distance (AD) to choose between routes from different routing sources. RIP has a default AD of 120, which is lower (more trustworthy) than OSPF's AD of 110 or EIGRP's AD of 90. The route with the lowest AD is chosen, regardless of its metric.

   EIGRP 0